Coherent memory norms on central charge windows Input tails and a strong quadratic-core expansion

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Abstract

We retain the coherence between adjacent charge sectors and derive a general-charge formula for the second-chaos fermionic-force norm. Its leading central-window limit agrees with half filling, although the intermediate contiguous relation changes. Factorial input-rank tails and the previous sixth output moment then prove a uniform strong replacement of the memory filter. The quadratic memory core can be assembled with the charge-dependent conserved correction. The output-deficit cross entries and full dynamical Gram still require a separate central-window calculation.

1 State and the actual operator

Use standard CAR, the full overlapping complex Gaussian p-body Hamiltonian, normalized maximally mixed trace at charge m, and annealed Hilbert–Schmidt norm, with $\hbar = 1$. Write

$$N = p^2/\lambda, \quad m = N/2 + xp, \quad n = N - m + 1, \quad d = m - p, \quad q = e^{-4\lambda}. \tag{1}$$

Here $|x| \le L$ and λ lies in a positive compact set. All estimates are uniform on these sets, with integer charges; actual $\lambda_N = p^2/N$ can be used throughout. The same coupling realization and variance w_N are used in $h = H_m$ and $k = H_{m-1}$. Define

$$\operatorname{Rem} X = \sum_i c_i^\dagger X c_i, \quad A = \operatorname{Rem}(k^2) + (2p - m)h^2, \quad O = \mathsf{W}_2 A, \quad B = \mathsf{W}_2 h^2 = h^2 - C_m I, \tag{2}$$

where $C_m = \mathbb{E}\operatorname{tr}_m h^2 = w_N \binom{m}{p}\binom{N-m+p}{p}$. The notation W_2 denotes ordinary Gaussian second chaos. The raw quartic core used below is $\mathcal{E} = \sum_i ([H,[H,c_i]])^\dagger [H,[H,c_i]] + [h,[h,A]]/4$, with full-Fock operations performed before restriction. For finite normalized C_m, the central result is

$$\boxed{\|O\|_2^2/C_m^2 \longrightarrow 4\lambda + 4\lambda^2(1+q), \qquad \langle B, O\rangle/C_m^2 \longrightarrow 2\lambda(1-q).} \tag{3}$$

Thus the leading force fluctuation survives without a new charge-dependent variance in this compact central window. The sector-dependent conserved subtraction from the previous note must nevertheless be retained.

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2 General-charge coherence and the finite norm

Let h_a, k_a be coherently embedded rank-a components, $C_a = \mathbb{E}h_a^2$, and $\omega_a = C_a/C_m$. Put $x_a = a(N+1-a)$. The established charged-covariance framework [1, 2] gives

$$R_a := C_a^-/C_a = \frac{d^2 n}{m(mn - x_a)}, \qquad \|\operatorname{Rem} X_r\|_{2,m}^2 = \frac{m}{n}(mn - x_r)\|X_r\|_{2,m-1}^2. \tag{4}$$

The latter follows directly from the adjacent-charge norm ratio of a disjoint rank-r monomial and equivariance. It is at most $m^2\|X_r\|^2$, and R_a is bounded uniformly in the central window.

Let $\chi = \chi_{ab}^{(m)}$ be the neutral Racah kernel and define

$$\xi = {}_4F_3\left[\begin{matrix} -a, a-N-1, -b, b-N-1 \\ 1-m, m-N, -N-1 \end{matrix}; 1\right], \quad D = \xi - \chi. \tag{5}$$

For the charged kernel $T_a(t)$, whose upper entries are $-a, a-N-1, -t, t-N$ and lower entries are those of ξ, the two charged components of a neutral tensor yield

$$\xi = T_a(b) - \frac{b}{N+1-2b}[T_a(b-1) - T_a(b)]. \tag{6}$$

This identity can also be checked term by term in the terminating series; it requires no half-filled product-parity rule. The direct and exchanged coherent contractions, per ordered rank pair and divided by C_aC_b, are g and $g\xi$, respectively, where

$$g = d^2/m, \quad h_0 = 2p - m, \quad \eta = p^2/m = g + h_0. \tag{7}$$

For a summand c_j of χ, $\xi_j = mc_j/(m-j)$, whereas $(\chi^-)_j = m(n-j)c_j/[n(m-j)]$. Hence the exact relation is

$$\boxed{\chi^- := \chi_{ab}^{(m-1)} = \chi + \frac{n-m}{n}D.} \tag{8}$$

At half filling $n - m = 1$; on a central window it can be of order p.

Set $P = g^2 x_a x_b/[(mn - x_a)(mn - x_b)]$ and $C_f^- = (m-1)n(N+1)/N$. The ordered and exchanged Casimir product moments give the removed symmetric-product norm

$$R_a R_b \frac{m}{n}\left[(mn - x_a - x_b)(1+\chi^-) + \frac{x_a x_b}{C_f^-}\right]. \tag{9}$$

Adding $2h_0 g(1+\xi) + h_0^2(1+\chi)$ and using (8) gives the nonnegative pair coefficient

$$\begin{aligned} f_{ab} &= \eta^2(1+\chi) + P(mn/C_f^- - 1 - \chi) + KD, \\ K &= (g^2 - P)(n-m)/n + 2h_0 g, \qquad \|O\|^2/C_m^2 = \sum_{a,b}\omega_a\omega_b f_{ab}. \end{aligned} \tag{10}$$

The corresponding B cross is $b_{ab} = \eta(1+\chi) + gD$. Gaussian decoupling includes equal input ranks in this ordered sum.

3 Uniform cancellation of the apparently divergent terms

Write $a = p - u$, $b = p - v$. For fixed u, v, j, x, the exact neutral Racah summand has expansion

$$c_j = \frac{(-4\lambda)^j}{j!}\left[1 + \frac{L_1}{p} + \frac{L_2 + L_1^2/2}{p^2} + O(p^{-3})\right], \tag{11}$$

where

$$L_1 = -j(u + v + 2\lambda) - j(j-1), \tag{12}$$

$$L_2 = -\tfrac{j}{2}(u^2 + v^2) - \tfrac{1}{2}j(j-1)(u+v) - \tfrac{1}{6}j(j-1)(2j-1) + \lambda j(u+v+1) + \tfrac{3}{2}\lambda j(j-1) - \lambda^2 j + 4\lambda^2 x^2 j. \tag{13}$$

The $4\lambda^2 x^2 j$ term is new and is needed for the finite cancellation. Express (10) as an unsigned part plus a single terminating series:

$$f_{ab} = \eta^2 + P(mn/C_f^- - 1) + \sum_j c_j \left[\eta^2 - P + K\frac{j}{m-j}\right]. \tag{14}$$

The bracket begins at order p^2, with coefficient $-(j + 4\lambda)/\lambda$. Expand the bracket through order p^0 and multiply by (13). Summing powers of j by $(z\partial_z)^k e^z$ at $z = -4\lambda$ gives, for the p^2, p, p^0 coefficients of the series,

$$(0,\ 0,\ 4\lambda^2 q). \tag{15}$$

The unsigned part tends to $4\lambda(1 + \lambda)$. These are polynomial identities in u, v, x, λ, saved with the exact expansions. Thus $f_{p-u,p-v}$ has the limit in (3), independent of fixed u, v, x. Likewise $mD \to -4\lambda q$ gives $b_{ab} \to 2\lambda(1 - q)$.

To justify the sums, choose $K_p = \lceil \log p \rceil$. The exact summand ratio gives $|c_j| \le A^j/j!$ uniformly for $a, b \le p$ and bounded x. Also $m - j$ and the scaled rational denominators stay away from zero. The bracket in (14) is bounded by $Bp^2(1 + j)$. Hence the $j > K_p$ contribution is superpolynomially small. On $u, v, j \le K_p$, logarithmic Taylor remainders are $O((1+u+v+j)^d/p^3)$ uniformly in x. After the p^2 multiplication, the residual sum is $O((\log p)^{d'}/p)$, which tends to zero. This proves convergence uniformly on the growing logarithmic input window.

4 The input tail and a three-direction Gram block

The exact input-rank weight ratio, with $\nu = p-u$, $a_\nu = \nu/(N-\nu+1)$ and $b_\nu = (\nu-1)/(N-\nu+2)$, is

$$\frac{\omega_{\nu-1}}{\omega_\nu} = a_\nu^2 \frac{1-b_\nu^2}{1-a_\nu^2} \frac{m-\nu+1}{p-\nu+1} \frac{N-p-\nu+1}{N-m-\nu+1}. \tag{16}$$

Uniformly on the central window it is at most $A/(u+1)$; for fixed u it tends to $\lambda/(u+1)$. Normalization then gives a Poisson(λ) limit for the input deficit and a factorial envelope. Equation (4), symmetric-product bounds, and Cauchy–Schwarz give the crude positive estimate $0 \le f_{ab} \le Bp^4$ at all ranks. The factorial tail beats this power outside $u, v \le K_p$. Any fixed polynomial weight in u, v is allowed. In particular

$$\sum_{a\le b,r} (1+u+v)^2 \|\mathsf{W}_2 A_{ab;r}\|^2 \le K C_m^2. \tag{17}$$

This is the needed bound on the actual coherent operator, rather than two separately estimated order-m pieces.

Let $D_N = \sum_a (p-a)h_a$, $\bar{u} = \sum_a (p-a)\omega_a$, and

$$L_{\mathrm{in}} = \mathsf{W}_2\{h, D_N - \bar{u}h\}. \tag{18}$$

This is an input-rank field; it is not the earlier cubic memory coefficient named L. The preceding limits and weighted tails give

$$\frac{\mathrm{Gram}(B, L_{\mathrm{in}}, O)}{C_m^2} \longrightarrow \begin{pmatrix} 1+q & 0 & 2\lambda(1-q) \\ 0 & 2\lambda(1+q) & 0 \\ 2\lambda(1-q) & 0 & 4\lambda + 4\lambda^2(1+q) \end{pmatrix}. \tag{19}$$

For example the L_{in} weight is $u+v-2\bar{u}$; its squared expectation tends to 2λ, while both its mean and its cross with the asymptotically constant b_{ab} vanish. The matrix is positive definite for $\lambda > 0$. Subtracting the exact finite-N projection on B from O gives

$$\|Y\|^2/C_m^2 \longrightarrow 4\lambda + \frac{16\lambda^2 q}{1+q} > 0. \tag{20}$$

The positive weighted pair norm also has the explicit limit

$$\sum_{a,b} (1+u+v)^2 \omega_a \omega_b f_{ab} \longrightarrow [4\lambda + 4\lambda^2(1+q)][(1+2\lambda)^2 + 2\lambda]. \tag{21}$$

5 A strong filter replacement and the assembled core

Use $X \odot Y = \{X, Y\}$ for distinct input ranks and squares for equal ranks. For $Z^{+}_{ab;r} = P_r(h_a \odot h_b)$, $Z^{-}_{ab;r} = P_r \operatorname{Rem}(k_a \odot k_b)$, put $A_{ab;r} = Z^{-}_{ab;r} + (2p - m)Z^{+}_{ab;r}$ and $S = a + b - r$. The previous uniform sixth moment and (4) imply

$$\sum_{a \le b,\ S > R} \|\mathsf{W}_2 A_{ab;r}\|^2 \le K C_m^2 p^4 R^{-6}. \tag{22}$$

Here separate charge bounds are used only outside the growing output window, retaining the full factor p^4. For the exact two-charge filter, the already verified central rational coefficients and binomial finite differences give

$$|g_A/C_m| \le K, \qquad |g_A/C_m - \gamma| \le K(1 + u + v + S)/p, \qquad \gamma = (3 - 4q + q^2)/2. \tag{23}$$

Orthogonality of input-pair chaos blocks and output ranks, (17), and (22) give

$$\Big\| \sum (g_A - C_m\gamma)\mathsf{W}_2 A_{ab;r} \Big\|^2 \le K C_m^4 [p^{-2} + R^2/p^2 + p^4/R^6]. \tag{24}$$

Taking $R = p^{3/4}$ proves, uniformly in the central window,

$$\boxed{\sum g_A \mathsf{W}_2 A_{ab;r} = C_m \gamma O + O_{L^2}(p^{-1/4}).} \tag{25}$$

The H^2 part can now be included. At arbitrary rank proportions $(a, b, r)/p = (\alpha, \beta, \rho)$ its rational prefactors have finite parts

$$\lambda(1 + 4x,\ 6 - 11\alpha - 4x,\ 6 - 11\beta - 4x, 5\alpha + 5\beta + 6\rho - 13 + 4x), \tag{26}$$

following the leading $(2, -2, -2, 2)p$. Consequently the centered g_H filter is bounded by $K(1 + u + v + S)$ uniformly. Its fixed-deficit limit from the previous note passes through the rank sum by factorial input tails and the sixth output moment. With $\mathcal{S}_N = \sum S \mathsf{W}_2 Z^{+}_{ab;r}$, the full second chaos is

$$\begin{aligned} \mathsf{W}_2 \mathcal{E} = {} & C_m\{2p(1 - q)^2 + \bar{K} + 4\lambda x(1 - q)^2\}B + C_m \gamma O \\ & + 8\lambda C_m(q^2 - q)L_{\mathrm{in}} + 8\lambda C_m q^2 \mathcal{S}_N + o_{L^2}(1), \end{aligned} \tag{27}$$

$$\bar{K} = \lambda(1 - 10q + 9q^2) - 16\lambda^2 q^2. \tag{28}$$

Use actual C_m, λ_N, q_N, x_N in the growing subtraction. The whole error is stated as $o_{L^2}(1)$; the rate in (25) applies specifically to its A-filter part.

6 Executed verification and the remaining spectral step

The general-charge formula is checked in 1331 exact rational rank pairs, including the contiguous relation, positivity, and equality of raw and cancellation-organized norms. Four independent full CAR systems $(N, p, m) = (5, 1, 2), (5, 2, 3), (6, 1, 4), (6, 2, 4)$ verify the (B, O) Gram using all 31466 symmetric Gaussian coordinate pairs. They preserve the same couplings in both charges.

The central symbolic calculation verifies the p^2 and p cancellations and the finite term as identities in x, u, v, λ. The input-weight ratio and general-rank core prefactors are checked symbolically. The numerical implementation sums all $p + 1$ input ranks and the complete terminating Racah series, with no rank cutoff. It compares 80 and 120 digits in 36 cases: $\lambda = 0.1, 0.25, 0.5$, $p = 16, 32, 64, 128$, and $x = -2, 0, 2$. The nine Gram entries and weighted A norm agree to all 65 stored digits. Four finite exact (B, O) matrices additionally validate the factorized summation.

For the last case, $\lambda = 0.5, p = 128, x = 2$, the finite $\|O\|^2/C_m^2 = 3.07899090506715$, compared with the limit 3.13533528323661; the weighted norm is 15.6459435799004, compared with 15.6766764161831. The finite relative errors are about -1.80% and -0.196%. Their cancellation factors are approximately 5855 and 5935. These discrepancies are retained; precision agreement is not convergence to the limit or interval certification.

The four scientific JSON records retain input hashes, parameters, versions, and diagnostics. The initial symbolic adaptation exposed a name collision and a half-filling-only constant assumption; both were corrected with their original sources retained. An inherited half-filled four-field Gram was removed from the central record to keep the scope explicit. This note establishes (19) and (27), not that four-field Gram at general charge.

Next compute the output-deficit entries, especially $\langle \mathcal{S}_N, O \rangle$, without the exact half-filled product-parity shortcut. Then extend energy insertions and the raw memory spectrum, control the charge-distribution tail and the two-insertion finite-memory remainder, and return to the Susskind singlet/meson spatial and mass dictionary.

References

[1] V. K. B. Kota and M. Vyas, arXiv:1504.00759, Sec. 7.1: finite particle-removal moments and rectangular sector maps. Its independent transition operator is distinct from the same-H force insertion studied here.

[2] A. Day and Chengheng, local notes: *A coherent covariance transform*; *The quadratic memory core in the double-scaled limit*; *Central charge windows for the memory calculation* (2026-11-06).